Compute Bipartite Network Metrics from Community Data

Description

Extracts the binary community matrix from a 'data_to_fit' list, binarizes it (any value > 0 becomes 1), and computes whole-network metrics via 'bipartite::networklevel()'. Returns a tidy tibble with one row per metric.

Usage

```
compute_network_metrics(

  data_to_fit = NULL,

  vec_indices = c("connectance", "nestedness", "modularity")

)
```

Arguments

`data_to_fit`	A named list as returned by 'build_jsdm_fit_input()'. Must contain the element 'data_community_to_fit', a numeric matrix with samples as rows and taxa as columns.
`vec_indices`	Character vector of network-level indices to compute, passed directly to the 'index' argument of 'bipartite::networklevel()'. Default: 'c("connectance", "nestedness", "modularity")'.

Details

The function treats the community matrix as a lower-level (samples) × upper-level (taxa) bipartite network. Only the binary incidence (presence/absence) is used: all values > 0 are set to 1 before passing to 'bipartite::networklevel()'.

If the binarized matrix has no positive entries (i.e. no species observed in any sample), an error is raised.

Value

A 'tibble' with two columns:

'metric'

: Character. Name of the network metric.

'value'

: Numeric. Computed value of the metric.

See Also

[build_jsdm_fit_input()], [compute_community_presence_absence()]